

Linear Algebra

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1 Linear Algebra Through Matrices

The prerequisites for this course include a sound knowledge of matrices up to the level of high school. This also entails some basic knowledge of linear algebra including solving linear systems of equations by hand, possibly up to 3 variables.

Knowledge of set theory is crucial, too. We also make the remark that for the rest of this course, all vectors shall be treated as column vectors and consequently multiplication of matrices will be done from the left side. That is, if a matrix A acts on a vector v , then the product is Av and not $v^\top A$.

Definition 1.1. A map $f : \mathbb{C}^n \rightarrow \mathbb{C}$ is said to be a **linear map** over $\mathbb{C}$ if

$$f(\alpha x + \beta y) = \alpha f(x) + \beta f(y)$$

for every $x, y \in \mathbb{C}^n$ and $\alpha, \beta \in \mathbb{C}$.

Linear maps over $\mathbb{C}$ can be structured nicely as follows.

Proposition 1.2. *Let $f : \mathbb{C}^n \rightarrow \mathbb{C}$ be a linear map over $\mathbb{C}$. Then there exists some $\alpha_1, \dots, \alpha_n \in \mathbb{C}$ such that*

$$f(x) = f([x_1, \dots, x_n]^\top) = \sum_{i=1}^n \alpha_i x_i$$

Conversely, every map of the above form is linear.

Proof. The converse is trivial to verify. To prove the forward direction we write our vector x as

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix} + \dots + x_n \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

Since the map f is linear we have

$$f(x) = x_1 f(e_1) + \dots + x_n f(e_n)$$

where for $1 \leq i \leq n$, e_i is the vector in $\mathbb{C}^n$ with 1 in the i -th place and 0 elsewhere. These $f(e_i) \in \mathbb{C}$ are the required values of α_i , that is, we set $\alpha_i = f(e_i)$ to get the required form. ■

The above proposition allows us to completely characterise maps $f : \mathbb{C}^n \rightarrow \mathbb{C}^m$ which are $\mathbb{C}$ -linear in each coordinate. The resulting structural coefficients are essentially the information a matrix stores.

Since linear algebra through matrices is used to deal with solving systems of equations, we define the following matrix which will be central to most of our techniques for solving such systems.

Definition 1.3. Fix $m, n \geq 1$ and a field $\mathbb{F}$. A **linear system** (over $\mathbb{F}$) of m equations in n variables is written completely as $Ax = b$ where A is an $m \times n$ matrix, x is an unknown vector of length n and b is a known vector of length m . All the entries are taken from the same field $\mathbb{F}$.

We write $A^+ = [A \mid b]$ and call it the **augmented matrix** associated with the system.

Those who do not know what a field is (the formal definition of which you will soon encounter at a later point in the notes) may consider $\mathbb{F} = \mathbb{C}$ and always think of everything going on over the complex numbers. Real numbers also do form a field and one may also use the real numbers to think of fields at an early stage. Of course, real numbers lack the property of ‘being closed’ since $x^2 + 1 = 0$ has no roots over the reals but any polynomial with possibly complex coefficients always has all roots being complex numbers and is ‘closed’ in some sense.

We now describe some fundamental operations that may be performed on systems which preserve the solutions.

Proposition 1.4. *The following operations on a system of linear equations give a new system of linear equations which furnishes the same solution as the original one.*

- (i) *Interchanging two equations.*
- (ii) *Multiplying an entire equation with a non-zero scalar from $\mathbb{F}$.*
- (iii) *Multiplying an equation by a scalar from $\mathbb{F}$ and adding it to another equation.*

Proof. We prove each of these by writing out the system and showing in both directions that if we have a solution of one system, then it is also a solution of the other system.

(i) Consider the following systems of equations:

$$\begin{array}{ccc}
\sum_{k=1}^n a_{1,k}x_k = b_1 & & \sum_{k=1}^n a_{1,k}x_k = b_1 \\
\vdots & & \vdots \\
\sum_{k=1}^n a_{i,k}x_k = b_i & & \sum_{k=1}^n a_{j,k}x_k = b_j \\
\vdots & & \vdots \\
\sum_{k=1}^n a_{j,k}x_k = b_j & & \sum_{k=1}^n a_{i,k}x_k = b_i \\
\vdots & & \vdots \\
\sum_{k=1}^n a_{m,k}x_k = b_m & & \sum_{k=1}^n a_{m,k}x_k = b_m \\
(S_1) & & (S_2)
\end{array}$$

Suppose $(\alpha_1, \dots, \alpha_n)$ satisfies system S_1 then it is clear that the same also satisfies S_2 and conversely if a solution satisfies S_2 then it also satisfies S_1 . Thus, S_1 and S_2 have the same solution sets.

(ii) Let $c \in \mathbb{F} \setminus \{0\}$ be a non-zero scalar. Consider the following systems of equations:

$$\begin{array}{ccc}
\sum_{k=1}^n a_{1,k}x_k = b_1 & & \sum_{k=1}^n a_{1,k}x_k = b_1 \\
\vdots & & \vdots \\
\sum_{k=1}^n a_{i,k}x_k = b_i & & \sum_{k=1}^n (ca_{i,k})x_k = cb_i \\
\vdots & & \vdots \\
\sum_{k=1}^n a_{m,k}x_k = b_m & & \sum_{k=1}^n a_{m,k}x_k = b_m \\
(S_1) & & (S_2)
\end{array}$$

Suppose $(\alpha_1, \dots, \alpha_n)$ satisfies system S_1 . Then the i -th equation holds, meaning $\sum_{k=1}^n a_{i,k}\alpha_k = b_i$. Multiplying both sides by the scalar c yields $\sum_{k=1}^n (ca_{i,k})\alpha_k = cb_i$, which implies the solution satisfies S_2 . Conversely, suppose a solution satisfies S_2 . Since $c \neq 0$, the multiplicative inverse c^{-1} exists in $\mathbb{F}$. Multiplying the i -th equation of S_2 by c^{-1} recovers the i -th equation of S_1 . Thus, S_1 and S_2 have the same solution sets.

(iii) Let $c \in \mathbb{F}$ be a scalar. We multiply the i -th equation by c and add it to the j -th equation. Consider

the systems:

$$\begin{array}{ccc}
\sum_{k=1}^n a_{1,k}x_k = b_1 & & \sum_{k=1}^n a_{1,k}x_k = b_1 \\
\vdots & & \vdots \\
\sum_{k=1}^n a_{i,k}x_k = b_i & & \sum_{k=1}^n a_{i,k}x_k = b_i \\
\vdots & & \vdots \\
\sum_{k=1}^n a_{j,k}x_k = b_j & & \sum_{k=1}^n (a_{j,k} + ca_{i,k})x_k = b_j + cb_i \\
\vdots & & \vdots \\
\sum_{k=1}^n a_{m,k}x_k = b_m & & \sum_{k=1}^n a_{m,k}x_k = b_m
\end{array}$$

(S_1)
 (S_2)

Suppose $(\alpha_1, \dots, \alpha_n)$ satisfies system S_1 . Then both $\sum_{k=1}^n a_{i,k}\alpha_k = b_i$ and $\sum_{k=1}^n a_{j,k}\alpha_k = b_j$ hold. Multiplying the former by c and adding it to the latter gives $\sum_{k=1}^n a_{j,k}\alpha_k + c \sum_{k=1}^n a_{i,k}\alpha_k = b_j + cb_i$, which simplifies to $\sum_{k=1}^n (a_{j,k} + ca_{i,k})\alpha_k = b_j + cb_i$. This is exactly the j -th equation in S_2 , so the solution satisfies S_2 . Conversely, if a solution satisfies S_2 , it satisfies both the i -th equation and the modified j -th equation. Subtracting c times the i -th equation from the j -th equation of S_2 yields $\sum_{k=1}^n (a_{j,k} + ca_{i,k})\alpha_k - c \sum_{k=1}^n a_{i,k}\alpha_k = b_j + cb_i - cb_i$, which simplifies back to $\sum_{k=1}^n a_{j,k}\alpha_k = b_j$. Therefore, the solution also satisfies S_1 . ■

Performing operations algorithmically can be described in plain text easily. However, we need to formalise it into the language of matrices in order to be able to do something meaningful with them. To this extent we define the following matrices which we connect to the above proposition in a proposition below.

Definition 1.5. Given a field $\mathbb{F}$ and some positive integer $n \geq 1$, for $1 \leq i, j \leq n$ we define the following three matrices, known as **elementary row matrices**.

$$\begin{aligned}
P_{i,j} &= I - e_i e_i^\top - e_j e_j^\top + e_i e_j^\top + e_j e_i^\top \\
M_j(c) &= \text{Diag}[1, \dots, c, \dots, 1] \quad (c \text{ in the } j\text{-th position}) \\
E_{i,j}(c) &= I + c e_i e_j^\top
\end{aligned}$$

Here e_i is the vector of length n over $\mathbb{F}$ with 1 in the i -th entry and 0 elsewhere.

One may notice that performing the three operations from Proposition 1.4 on the identity matrix I gives us the three respective matrices defined above. This is indeed the general idea. Instead of asking how new matrix would look when the operations are performed on a matrix A , we ask how it would look when performed on $IA = A$. Using some magical associativity law (think about this), we may apply the operations on I and then the new matrix is easy to describe as defined above.

Proposition 1.6. *In accordance with the above Proposition 1.4, each of the mentioned operations on the system of equations corresponds to the action of matrices on the augmented matrix as follows (assume standard parameters of m and n , that is, the system is $Ax = b$ with A being an $m \times n$ matrix over $\mathbb{F}$):*

- (i) *Interchanging the i -th and j -th equations is equivalent to left multiplying the augmented system by the $m \times m$ matrix $P_{i,j}$.*

- (ii) Multiplying the j -th equation by some non-zero scalar c is equivalent to left multiplying the augmented system by $M_j(c)$.
- (iii) Multiplying the j -th equation by a scalar c and adding it to the i -th equation is equivalent to left multiplying the augmented matrix by $E_{i,j}(c)$.

Proof. Suppose the augmented matrix is some $m \times q$ matrix B (where $q = n + 1$), which we can represent in terms of its rows as:

$$B = \begin{bmatrix} b_{1,1} & b_{1,2} & \cdots & b_{1,q} \\ b_{2,1} & b_{2,2} & \cdots & b_{2,q} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m,1} & b_{m,2} & \cdots & b_{m,q} \end{bmatrix} = \begin{bmatrix} R_1 \\ R_2 \\ \vdots \\ R_m \end{bmatrix}$$

where each $R_k = [b_{k,1}, b_{k,2}, \dots, b_{k,q}]$ is a row vector of length q . Observe that left-multiplying B by the vector $e_k^\top$ extracts the k -th row of B , meaning $e_k^\top B = R_k$. Furthermore, multiplying a column vector e_k by the row vector R_l yields an $m \times q$ matrix with R_l in the k -th row and zeros everywhere else. We now evaluate the action of each matrix:

- (i) For the permutation matrix $P_{i,j}$, we have:

$$\begin{aligned} P_{i,j}B &= (I - e_i e_i^\top - e_j e_j^\top + e_i e_j^\top + e_j e_i^\top)B \\ &= B - e_i(e_i^\top B) - e_j(e_j^\top B) + e_i(e_j^\top B) + e_j(e_i^\top B) \\ &= B - e_i R_i - e_j R_j + e_i R_j + e_j R_i \end{aligned}$$

Subtracting $e_i R_i$ and $e_j R_j$ from B replaces the i -th and j -th rows with zeros. Adding $e_i R_j$ and $e_j R_i$ subsequently places the old j -th row into the i -th position, and the old i -th row into the j -th position. This exactly interchanges the two equations.

- (ii) For the scaling matrix $M_j(c)$, we can rewrite it as $M_j(c) = I + (c - 1)e_j e_j^\top$. Thus:

$$\begin{aligned} M_j(c)B &= (I + (c - 1)e_j e_j^\top)B \\ &= B + (c - 1)e_j(e_j^\top B) \\ &= B + (c - 1)e_j R_j \end{aligned}$$

This operation adds $(c - 1)$ times the j -th row to the existing j -th row in B . The new j -th row becomes $R_j + (c - 1)R_j = cR_j$, leaving all other rows unaffected. This matches multiplying the j -th equation by the non-zero scalar c .

- (iii) For the matrix $E_{i,j}(c)$, we have:

$$\begin{aligned} E_{i,j}(c)B &= (I + c e_i e_j^\top)B \\ &= IB + c e_i(e_j^\top B) \\ &= B + c e_i R_j \end{aligned}$$

The matrix $c e_i R_j$ consists of $c R_j$ in its i -th row and zeros elsewhere. Adding this to B updates the i -th row from R_i to $R_i + c R_j$, which is precisely the operation of multiplying the j -th equation by c and adding it to the i -th equation. ■

Proposition 1.7. *The operation of interchanging two equations can be achieved by a sequence of the other two operations.*

Proof. The following is a one line proof of the above proposition.

$$P_{i,j} = E_{i,j}(1)E_{j,i}(-1)M_i(-1)E_{i,j}(-1)$$

One is encouraged to verify the above directly in terms of the equations themselves. ■

We now define the notion of a system being in row echelon form, which is a very convenient form to have for solving systems of equations.

Definition 1.8. A matrix is said to be in **row echelon form** if the following conditions hold:

- (i) If there is a row that consists entirely of zeros, then it is at the bottom of the system.
- (ii) For every non-zero row, the first non-zero entry (called the leading coefficient) is strictly to the right of the leading coefficient of the row above it.

The first non-zero entry in a non-zero row is called a **pivot**. The column in which a pivot appears is called a **pivot column**. The other columns are called **free columns**.

We define a stricter form of row echelon form, called reduced row echelon form, which is even more convenient to work with, as we shall see.

Definition 1.9. A matrix is said to be in **reduced row echelon form** if it is in row echelon form and the following additional conditions hold:

- (i) Every pivot is 1.
- (ii) Each pivot is the only non-zero entry in its column.

Theorem 1.10. *The row reduced echelon form of a matrix is unique.*

Proof. The proof is by induction on the number of columns. Suppose A has a single column. Then any REF is necessarily of the form

$$\begin{bmatrix} \alpha \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

by the definition of REF. If $\alpha = 0$, then the RREF is the same as the REF. If $\alpha \neq 0$, then the RREF is

$$\begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Now suppose that the result holds for any $m \times n$ matrix. We prove it for matrices of size $m \times (n + 1)$. Let A be an $m \times (n + 1)$ matrix and let B and C be two RREFs of A .

Let A' be the $m \times n$ matrix obtained by deleting the last column of A , and let B' and C' be the $m \times n$ matrices obtained by deleting the last columns of B and C , respectively. Since B and C are row equivalent to A , there exist invertible matrices E and F (each a product of elementary row matrices) such that $EA = B$ and $FA = C$. This fact is later proved in Proposition 1.14.

By the definition of matrix multiplication by columns, the j -th column of B is simply E applied to the j -th column of A . Therefore, truncating the last column yields $EA' = B'$ and $FA' = C'$. This establishes that B' and C' are row equivalent to A' .

Furthermore, since B and C are in RREF, removing their final columns cannot violate any of the structural conditions for being in RREF (the pivots remain 1, the columns with pivots remain standard basis vectors, and the staircase structure is strictly preserved). Thus, B' and C' are RREFs of A' . By the induction hypothesis, we have $B' = C'$.

It remains to show that the last columns of B and C , say b and c , are also equal. Consider the matrix $B - C$. This matrix is zero everywhere except possibly in the last column which is $b - c$. Suppose this is non-zero.

Let $z \in \mathbb{C}^{n+1}$ be such that $Az = 0$. Applying ERMs does not change the solution set of the homogeneous system, so we have $Bz = 0$ and $Cz = 0$. This implies that $(B - C)z = 0$. Writing it out we get

$$\begin{bmatrix} 0 & 0 & \cdots & 0 & b_1 - c_1 \\ 0 & 0 & \cdots & 0 & b_2 - c_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & b_m - c_m \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_{n+1} \end{bmatrix} = \begin{bmatrix} (b_1 - c_1)z_{n+1} \\ (b_2 - c_2)z_{n+1} \\ \vdots \\ (b_m - c_m)z_{n+1} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Since some $b_j - c_j$ ($1 \leq j \leq m$) is non-zero, we have $z_{n+1} = 0$. This means that any vector in the null space of A has a zero in its last coordinate. Consequently, the $(n + 1)$ -th column of A cannot be expressed as a linear combination of the first n columns (think why).

Because elementary row operations preserve linear relations among columns, the last columns of B and C (which are b and c) cannot be linear combinations of the columns of B' and C' , respectively. By the definition of RREF, a column is not a linear combination of previous columns if and only if it introduces a new pivot.

Since $B' = C'$ by the induction hypothesis, they have the exact same number of pivots, say r . Therefore, the new pivot in both B and C must be the $(r + 1)$ -th standard basis vector. Hence $b = c = e_{r+1}$, which contradicts the assumption that $b - c$ is non-zero. Therefore, we must have $b = c$, which completes the inductive step. ■

Definition 1.11. A variable x_i involved in $[A \mid b]$ is called a **simple variable** if the i -th column of the row reduced matrix of A is a pivot column. Otherwise, it is called a **free variable**. Here, A is of size $m \times n$ and b is of length m . x is the unknown vector of length n .

Proposition 1.12. All simple variables x_{a_r} associated with a solvable system $[A \mid b]$ are expressible strictly in terms of constants and free variables x_j with $j > a_r$. Consequently, the free variables are the only variables that can be assigned arbitrary values in $\mathbb{C}$ to obtain a solution to the system.

Proof. Let $a_1 < a_2 < \cdots < a_k$ be the labels of the pivot columns of A_{RREF} . We wish to solve the augmented system $[A_{\text{RREF}} \mid c]$, where the row reduction $A \xrightarrow{\text{RREF}} A_{\text{RREF}}$ induces the transformation of the vector $b \xrightarrow{\text{RREF}} c$.

Clearly, the number of pivots cannot exceed the dimensions of the matrix, so $k \leq \min\{m, n\}$. In particular, $k \leq m$, where m is the number of rows. By the definition of row echelon form, the first k rows contain the pivots, and all rows below the k -th row must necessarily be zero rows. Since the system is solvable we must have $c_{k+1} = c_{k+2} = \cdots = c_m = 0$.

Let us describe the rows of $[A_{\text{RREF}} \mid c]$ in more detail. The first row is of the form $0 \cdots 0, 1, \alpha_{i_1}, \alpha_{i_2} \cdots \alpha_{i_{l_1}}, c_1$. The second row has the form $0 \cdots 0, 1, \beta_{i_1}, \beta_{i_2} \cdots \beta_{i_{l_2}}, c_2$. And so on until the last column which has the form $0 \cdots 0, 1, \gamma_{i_1}, \gamma_{i_2} \cdots \gamma_{i_{l_k}}, c_k$. Here, the 1 in the i -th row is located at column a_i and $l_i = n - a_i$ for $1 \leq i \leq k$.

Now, consider the equation corresponding to the r -th row for any $1 \leq r \leq k$. The leading non-zero entry is a pivot of 1 located at column a_r . By the second condition of the definition of reduced row echelon form, this pivot is the only non-zero entry in its column. Furthermore, for any other pivot column a_s (where $s \neq r$), the entry in the r -th row is forced to be 0.

Consequently, when we look at the r -th row, the coefficients corresponding to all other simple variables x_{a_s} are zeroes (they either lie above a pivot 1 or below it). Because the staircase structure of a row echelon form dictates that all entries to the left of the pivot column a_r must be zero, the only variables that can appear with non-zero coefficients in this row equation must have column indices j strictly greater than a_r and must be free variables.

Letting $u_{r,j}$ denote the scalar entry of A_{RREF} in the r -th row and j -th column, we can write out this row equation explicitly as:

$$1 \cdot x_{a_r} + \sum_{\substack{j > a_r \\ j \text{ is free}}} u_{r,j} x_j = c_r$$

Rearranging this equation allows us to isolate the simple variable x_{a_r} :

$$x_{a_r} = c_r - \sum_{\substack{j > a_r \\ j \text{ is free}}} u_{r,j} x_j$$

This explicit formulation holds independently for each row $r \in \{1, \dots, k\}$. It demonstrates that the value of every simple variable is rigidly determined by the system's constants and the chosen values of the free variables. Consequently, one cannot assign arbitrary values to the simple variables; their values are entirely dependent on the choices made for the free variables.

Conversely, the free variables do not appear as leading variables in any equation of the reduced system, meaning they are entirely unconstrained. We may therefore assign them any arbitrary values in $\mathbb{C}$, and the above equations will uniquely dictate the necessary values for the simple variables to form a valid solution vector. This establishes that the free variables are indeed the only variables that afford such arbitrary assignment. ■

We now make some statements on the relationship between the number of pivots and the number of solutions of the system.

Proposition 1.13. *Let A be in row reduced echelon form. If every row has a pivot then the system $[A \mid b]$ has at least one solution. If every column has a pivot then the system $[A \mid b]$ has at most one solution. If every row and every column has a pivot then the system $[A \mid b]$ has exactly one solution, A is square and in fact $A = I$.*

Proof. Let A be of size $m \times n$. If every row has a pivot, then $m \leq n$. If every column has a pivot, then $n \leq m$. Thus, if every row and every column has a pivot, we must have $m = n$. Since A is in RREF and has pivots in every row and every column, the only possibility is that A is the identity matrix I . In this case, the system $[I \mid b]$ has exactly one solution, which is $x = b$.

Now suppose every row has a pivot. Then $m \leq n$. This means that there are m simple variables and $n - m$ free variables. Since the system is consistent, we can assign arbitrary values to the free variables and solve for the simple variables using the equations corresponding to the pivot rows. This guarantees at least one solution.

Now suppose every column has a pivot. Then $n \leq m$. This means that there are n simple variables and no free variables. The values of the simple variables are completely determined by the equations corresponding to the pivot rows, and there are no free variables to assign arbitrary values to. Therefore, there can be at most one solution. In fact, since every column is a pivot column, we must have

$$A = \begin{bmatrix} I_{n \times n} \\ 0_{(m-n) \times n} \end{bmatrix}$$

■

We now prove that the RREF can be computed and obtained only using elementary row operations.

Proposition 1.14. *For any matrix A , there exists a finite sequence of elementary row operations that transforms A into its row reduced echelon form.*

Proof. We describe an algorithm to compute the RREF of A using elementary row operations. Let A be an $m \times n$ matrix. We start with the first column and look for the first non-zero entry in this column. If there is no non-zero entry, we move on to the second column and repeat this process until we find a column with a non-zero entry or we exhaust all columns.

Suppose we find a non-zero entry in the i -th row and j -th column. We can use the first operation (interchanging two equations) to swap the i -th row with the first row, bringing the non-zero entry to the top of the current submatrix. Next, we can use the second operation (scaling an equation) to multiply the first row

by the inverse of this non-zero entry, making it a pivot of 1. Finally, we can use the third operation (adding a multiple of one equation to another) to eliminate all other entries in the j -th column below this pivot by adding appropriate multiples of the first row to those rows.

After processing the first pivot column, we move on to the next column and repeat this process for the submatrix that excludes the rows above the current pivot row. We continue this process until we have processed all columns or all remaining rows are zero.

Once we have obtained a row echelon form, we can then perform back substitution using elementary row operations to ensure that each pivot is the only non-zero entry in its column, thus achieving reduced row echelon form.

Since each step involves a finite number of elementary row operations, and there are at most $\min(m, n)$ pivots to process, this algorithm will terminate after a finite number of steps, yielding the RREF of A . ■

Note that the above proof simply gives us the existence of a sequence of elementary row operations that transforms A into its RREF. There is no restriction on the RREF being unique. This avoids circularity since the uniqueness of RREF which was proved in Theorem 1.10 relied on the above proposition. However, the algorithm described in the above proof will always yield the unique RREF of A regardless of the choices made during the process, which is a consequence of the uniqueness of RREF.

The following is an immediate consequence of the above proposition.

Corollary 1.15. *Given a square matrix A , there exists a finite sequence of elementary row operations such that the resulting matrix is either the identity matrix I or has last row consisting of all zeroes.*

Proof. Let A be an $n \times n$ matrix. By Proposition 1.14, we can apply a finite sequence of elementary row operations to transform A into its RREF, say B . Since B is in RREF, it must have a staircase structure with pivots that are either 1 or absent. If B has pivots in every row and every column, then B must be the identity matrix I . If B does not have pivots in every row, then there must be at least one row that consists entirely of zeroes, and by the definition of row echelon form, this zero row must be at the bottom of the matrix. Thus, the resulting matrix is either the identity matrix or has its last row consisting of all zeroes. ■

We now visualise the RREF in a nice manner to understand the structure of the solution set of a system of equations. Here we need to allow the swapping of columns. Suppose A is an $m \times n$ matrix and has k pivots in its RREF. Then the first k rows of the RREF of A contain the pivots. And each pivot column has a single 1 in that pivot position. Thus, when we swap columns to bring the pivots together, the matrix takes the following form:

$$\begin{bmatrix} I_{r \times r} & F_{r \times p} \\ 0_{q \times r} & 0_{q \times p} \end{bmatrix}$$

Here, $p = n - r$ and $q = m - r$. The first r columns are the pivot columns and the last p columns are the free columns. The first r rows contain the pivots and the last q rows are zero rows. The matrix F contains the coefficients of the free variables in the equations corresponding to the pivot rows.

Motivated by the above corollary we make the following definition.

Definition 1.16. A square matrix A is said to be **invertible** if it can be transformed into the identity matrix I by a finite sequence of elementary row operations. If no such sequence exists, then A is said to be **non-invertible** or **singular**. If $E_1 \cdots E_m A = I$ then we call $E_1 \cdots E_m$ an inverse of A and denote it by $A^{-1} = E_1 \cdots E_m$.

Since matrices form a ring under addition and multiplication, the inverse is unique if it exists.

We now show that the elementary row matrices defined in Definition 1.5 are indeed invertible, which is a crucial fact that allows us to use them to perform row operations without worrying about losing invertibility. This is important because we want to be able to reverse the row operations if needed, and also because the

invertibility of these matrices ensures that they do not change the solution set of the system of equations when applied to the augmented matrix.

Proposition 1.17. *All the three elementary row matrices are invertible.*

Proof. Let E be an elementary row matrix. We need to show that there exists a matrix E^{-1} such that $EE^{-1} = E^{-1}E = I$. We simply provide the inverses. It is left to the reader to verify that these are indeed inverses by direct matrix multiplication.

- (i) The inverse of the permutation matrix $P_{i,j}$ is itself, since swapping the same two rows twice returns the original matrix. Thus, $P_{i,j}^{-1} = P_{i,j}$.
- (ii) The inverse of the scaling matrix $M_j(c)$ is $M_j(c^{-1})$, since multiplying the j -th row by c and then by c^{-1} returns the original row. Thus, $M_j(c)^{-1} = M_j(c^{-1})$.
- (iii) The inverse of the matrix $E_{i,j}(c)$ is $E_{i,j}(-c)$, since adding c times the j -th row to the i -th row and then adding $-c$ times the j -th row to the i -th row returns the original row. Thus, $E_{i,j}(c)^{-1} = E_{i,j}(-c)$. ■

Note that merely having the ERM's being invertible was not that important. What is important is that they are invertible by other ERM's, that is, the inverse of an ERM is also an ERM.

We end the first introductory section with a few equivalent conditions of invertibility.

Theorem 1.18. *Let A be an $n \times n$ matrix. The following are equivalent:*

- (i) A is invertible.
- (ii) The system $[A \mid b]$ has exactly one solution for every choice of b .
- (iii) A has n pivots in any REF.
- (iv) The RREF of A is the identity matrix I .
- (v) A is a product of elementary row matrices.

Proof. (1) $\implies$ (2): Suppose A is invertible. Let b be an arbitrary vector in $\mathbb{C}^n$. Consider the system $[A \mid b]$, which corresponds to the matrix equation $Ax = b$. Left-multiplying both sides by the inverse A^{-1} , we obtain $x = A^{-1}b$. This demonstrates that $x = A^{-1}b$ is a solution, so a solution exists. To show uniqueness, suppose y is another solution such that $Ay = b$. Left-multiplying by A^{-1} yields $y = A^{-1}b$. Thus, the solution is unique.

(2) $\implies$ (3): Suppose the system $[A \mid b]$ has exactly one solution for every choice of $b \in \mathbb{C}^n$. In particular, choosing $b = 0$, the homogeneous system $Ax = 0$ has exactly one solution. Since the trivial solution $x = 0$ always satisfies the homogeneous system, it must be the unique solution. A unique solution implies that there are no free variables in the system. Because the system has n total variables and no free variables, all n variables must be simple variables. By definition, every simple variable corresponds to a pivot column, meaning A must have exactly n pivots in any REF.

(3) $\implies$ (4): Follows from Proposition 1.13.

(4) $\implies$ (5): Suppose the RREF of A is the identity matrix I . Since the RREF is obtained by applying a sequence of elementary row operations, there exists a finite sequence of elementary row matrices $E_1, E_2, \dots, E_k$ such that $(E_k \cdots E_2 E_1)A = I$. Let $E = E_k \cdots E_2 E_1$. Then E is an invertible matrix and $EA = I$, which implies $A = E^{-1}$. Using the property of inverses of products, we have $A = (E_k \cdots E_2 E_1)^{-1} = E_1^{-1} E_2^{-1} \cdots E_k^{-1}$. Since the inverse of any elementary row matrix is itself an elementary row matrix, A is a product of elementary row matrices.

(5) $\implies$ (1): Suppose A is a product of elementary row matrices, say $A = E_1 E_2 \cdots E_m$. By definition, every elementary row matrix is invertible. Because the product of any finite number of invertible matrices is also invertible, it follows that A is invertible. ■

2 Vector Spaces and Subspaces

Before getting into the abstract definition of vector spaces, we first have to define the notion of a field, which is the underlying set over which vector spaces are defined.

Definition 2.1. A **field** is a set F with two operations $+$ and $\cdot$ such that:

- (i) $(F, +)$ is an abelian group with an identity element denoted 0 , that is,
 - (a) $+$ is associative: for all $a, b, c \in F$, we have $(a + b) + c = a + (b + c)$.
 - (b) $+$ is commutative: for all $a, b \in F$, we have $a + b = b + a$.
 - (c) There exists an identity element $0 \in F$ (called *zero*) such that for all $a \in F$, we have $a + 0 = a$.
 - (d) For every $a \in F$, there exists an inverse element $-a \in F$ such that $a + (-a) = 0$.
- (ii) $(F \setminus \{0\}, \cdot)$ is an abelian group with some identity element denoted 1 , that is,
 - (a) $\cdot$ is associative: for all $a, b, c \in F$, we have $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.
 - (b) $\cdot$ is commutative: for all $a, b \in F$, we have $a \cdot b = b \cdot a$.
 - (c) There exists an identity element $1 \in F$ (called *one*) such that for all $a \in F$, we have $a \cdot 1 = a$.
 - (d) For every non-zero $a \in F$, there exists an inverse element $a^{-1} \in F$ such that $a \cdot a^{-1} = 1$.
- (iii) The distributive property holds: for all $a, b, c \in F$, we have $a \cdot (b + c) = a \cdot b + a \cdot c$ and $(a + b) \cdot c = a \cdot c + b \cdot c$.

It is not very difficult to prove the uniqueness of both identities and the uniqueness of an inverse in both cases. One should prove this as an exercise. You are only allowed to use the above 9 axioms to prove this.

Some very popular examples of fields include $\mathbb{R}$ with the usual addition and multiplication, $\mathbb{C}$ with the usual addition and multiplication and $\mathbb{Z}/p\mathbb{Z}$ for some prime number p with modulo addition and modulo multiplication. For readers unfamiliar with $\mathbb{Z}/p\mathbb{Z}$ we clarify this in brief as follows. We define the set

$$\mathbb{Z}/p\mathbb{Z} = \{\bar{0}, \bar{1}, \dots, \overline{p-1}\}$$

and define $\bar{a} + \bar{b} = \overline{a + b}$ and $\bar{a} \cdot \bar{b} = \overline{a \cdot b}$. Here, we have the notion of $\bar{a} = \bar{b}$ if and only if $p \mid a - b$. As an example, for $p = 11$ we have

$$\bar{8} + \bar{9} = \overline{17} = \bar{6} \quad \bar{8} \cdot \bar{7} = \overline{56} = \bar{1}$$

Note that every non-zero element is invertible. As noted above, the inverse of $\bar{8}$ is $\bar{7}$ and vice versa.

Usually we work throughout with $\mathbb{R}$ or $\mathbb{C}$. To keep things as general as possible, we denote a general field by $\mathbb{F}$ and the reader may use $\mathbb{R}$ or $\mathbb{C}$ instead of this to understand things better.

We can now define a vector space over a given field as follows.

Definition 2.2. Let $\mathbb{F}$ be a field. By a **vector space** V over $\mathbb{F}$, we mean a set V equipped with a binary operation denoted by $+$ (called *vector addition*) and another operation $\phi : \mathbb{F} \times V \rightarrow V$ (called *scalar multiplication*) such that

- (i) $(V, +)$ is an abelian group with an identity element denoted 0 or $\vec{0}$, that is,
 - (a) $+$ is associative: for all $u, v, w \in V$, we have $(u + v) + w = u + (v + w)$.
 - (b) $+$ is commutative: for all $u, v \in V$, we have $u + v = v + u$.
 - (c) There exists an identity element $0 \in V$ (*zero vector*) such that for all $v \in V$, we have $v + 0 = v$.
 - (d) For every $v \in V$, there exists an inverse element $-v \in V$ such that $v + (-v) = 0$.
- (ii) The scalar multiplication, now denoted as $\lambda * v := \phi(\lambda, v)$ for $\lambda \in \mathbb{F}$ and $v \in V$, satisfies:
 - (a) $\lambda * (u + v) = \lambda * u + \lambda * v$ for all $\lambda \in \mathbb{F}$ and $u, v \in V$.
 - (b) $(\lambda + \mu) * v = \lambda * v + \mu * v$ for all $\lambda, \mu \in \mathbb{F}$ and $v \in V$.
 - (c) $(\lambda\mu) * v = \lambda * (\mu * v)$ for all $\lambda, \mu \in \mathbb{F}$ and $v \in V$.
 - (d) $1 * v = v$ for all $v \in V$, where 1 is the multiplicative identity in $\mathbb{F}$.

Note that if we want to specify a vector space, we must specify the field, the set of vectors, the addition of vectors and also scalar multiplication. Once these 4 are provided, one has to check the above 8 axioms to verify that the given set is indeed a vector space over the field.

We provide examples of vector spaces below. Verifying that they are indeed vector spaces is left to the reader.

- (i) **n -tuple space:** We consider some field $\mathbb{F}$ and positive integer n and define $V = \mathbb{F}^n$, that is, n -tuples where each coordinate comes from $\mathbb{F}$. Addition of vectors is component-wise and scalar multiplication is also component-wise. That is,

$$(x_1, \dots, x_n) + (y_1, \dots, y_n) := (x_1 + y_1, \dots, x_n + y_n)$$

$$\lambda * (x_1, \dots, x_n) = (\lambda x_1, \dots, \lambda x_n)$$

- (ii) **Matrix Spaces:** Fix positive integers m and n . We fix a field $\mathbb{F}$ and consider the set of all $m \times n$ matrices with entries from $\mathbb{F}$. We denote this set by $M_{m \times n}(\mathbb{F})$. Addition of matrices is defined as the component-wise addition of their entries (usual matrix addition), and scalar multiplication is defined as the component-wise multiplication of the matrix entries by the scalar (usual scaling of matrices by some constant).
- (iii) **Field over field:** Let $F \subseteq E$ be fields. Then E may be regarded as a vector space over F by using the usual addition $+_E$ of E to add vectors, that is, elements of E , and for $\lambda \in F$ and $v \in E$, we define $\lambda * v$ to be $\lambda \cdot_E v$, that is, the usual multiplication in E by treating $\lambda \in E$. Here, E is called a *field extension* of F . This gets us into field and Galois theory.
- (iv) **Polynomials over a field $\mathbb{F}$:** Fix a field $\mathbb{F}$ and an integer $n \geq -1$. Consider the space of all polynomials of degree at most n with coefficients from the field $\mathbb{F}$. Degree 0 polynomials are the non-zero constants while the zero polynomial is defined to be of degree -1 . Addition of polynomials is defined as the usual addition of polynomials, and scalar multiplication is defined as the usual scaling of polynomials by some constant. We denote this vector space by $\mathbb{F}[x]_{\leq n}$. If n is allowed to be as large as possible, we denote the set of all polynomials over $\mathbb{F}$ by $\mathbb{F}[x]$.

Note that the same set of vectors may form different vector space by changing the underlying field. For instance $\mathbb{C}$ is a vector space over both $\mathbb{C}$ and $\mathbb{R}$. The scalar multiplication is defined differently in each case. Later, we shall see that the former has dimension 1 while the latter has dimension 2.

Below we see some basic properties that follow from the 8 defining axioms.

Proposition 2.3. *Let V be a vector space over a field $\mathbb{F}$. Then for all $u, v \in V$ and $\lambda, \mu \in \mathbb{F}$, we have*

- (i) $0 * v = 0$.
- (ii) $\lambda * 0 = 0$.
- (iii) $(-\lambda) * v = -(\lambda * v)$.
- (iv) $\lambda * (-v) = -(\lambda * v)$.
- (v) If $\lambda * v = 0$, then either $\lambda = 0$ or $v = 0$.

Proof. We prove each part separately.

- (i) $0 * v = (0 + 0) * v = 0 * v + 0 * v$. By the cancellation law for abelian groups we have $0 * v = 0$.
- (ii) $\lambda * 0 = \lambda * (0 + 0) = \lambda * 0 + \lambda * 0$. Again, $\lambda * 0 = 0$.
- (iii) $(\lambda * v) + ((-\lambda) * v) = (\lambda + (-\lambda)) * v = 0 * v = 0$. Thus, $(-\lambda) * v$ is the additive inverse of $\lambda * v$, which is $-(\lambda * v)$.
- (iv) $\lambda * (-v) = \lambda * (-1 * v) = (-\lambda) * v$ by the previous part.
- (v) Suppose $\lambda * v = 0$. If $\lambda \neq 0$, then we can multiply both sides by λ^{-1} to get $v = \lambda^{-1} * (\lambda * v) = \lambda^{-1} * 0 = 0$. Thus, if $\lambda * v = 0$, then either $\lambda = 0$ or $v = 0$.

■

Henceforth we shall ignore the $*$ symbol and simply write λv instead of $\lambda * v$ for scalar multiplication. This is a common convention in linear algebra and should not cause any confusion.

We now define linear combinations, linear dependence and linear independence, which are fundamental concepts in the theory of vector spaces.

Definition 2.4. Let V be a vector space over a field $\mathbb{F}$ and let $v_1, \dots, v_n \in V$. A *linear combination* of $v_1, \dots, v_n$ is an element of the form

$$\lambda_1 v_1 + \dots + \lambda_n v_n$$

where $\lambda_1, \dots, \lambda_n \in \mathbb{F}$.

By the definition of a vector space, addition of vectors and scalar multiplication are binary operations and hence every linear combination of vectors is again a well-defined element of the vector space.

Definition 2.5. Let V be a vector space over a field $\mathbb{F}$ and let $v_1, \dots, v_n \in V$. The vectors $v_1, \dots, v_n$ are said to be **linearly independent** if the only linear combination that equals the zero vector is the trivial one where all coefficients are zero. That is,

$$\lambda_1 v_1 + \dots + \lambda_n v_n = 0 \iff \lambda_1 = \dots = \lambda_n = 0$$

Otherwise, the vectors are said to be **linearly dependent**.

We now define subspaces and linear spans.

Definition 2.6. Let V be a vector space over a field $\mathbb{F}$. A subset $W \subseteq V$ is called a **subspace** of V if itself is a vector space over $\mathbb{F}$ under the same addition and scalar multiplication as defined on V .

Instead of checking every axiom in the definition of a vector space to verify that a subset is a subspace, we have the following convenient criterion.

Proposition 2.7. Let V be a vector space over a field $\mathbb{F}$ and let $W \subseteq V$. Then W is a subspace of V if and only if the following conditions hold:

- (i) If $u, v \in W$, then $u + v \in W$.
- (ii) If $\lambda \in \mathbb{F}$ and $v \in W$, then $\lambda v \in W$.

Equivalently, W is a subspace of V if and only if for all $\lambda \in \mathbb{F}$ and $u, v \in W$, we have $\lambda u + \mu v \in W$.

Proof. The last statement about $\lambda u + \mu v$ is just a restatement of the second and third conditions. We first show that if W is a subspace of V , then the two conditions hold. Clearly, since W is a vector space, it is closed under addition and scalar multiplication, so the two conditions hold.

Conversely, suppose the two conditions hold. We need to show that W is a vector space under the same operations as V . Firstly, the operations of vector addition and scalar multiplication are well-defined on W since W is closed under these operations by assumption. We now check the eight axioms of a vector space for W .

- (i) Axioms of addition of vectors
 - (a) Associativity follows because the addition in W is the same as the addition in V , and V satisfies associativity.
 - (b) As above, commutativity is also true since it is inherited from V .
 - (c) Identity element: Since W is closed under scalar multiplication, we have $0 * v = 0 \in W$ for all $v \in W$. Thus, the zero vector of V is also the zero vector of W and serves as the identity element for addition in W .
 - (d) Inverse element: For every $v \in W$, we have $-1 * v = -v \in W$ since W is closed under scalar multiplication. Thus, every element in W has an additive inverse in W .
- (ii) Axioms of scalar multiplication

- (a) $\lambda * (u + v) = \lambda * u + \lambda * v$ holds in V and hence also in W .
- (b) $(\lambda + \mu) * v = \lambda * v + \mu * v$ also holds in V and hence in W .
- (c) $(\lambda\mu) * v = \lambda * (\mu * v)$ holds in V and hence in W .
- (d) $1 * v = v$ holds in V and hence in W .

Since W satisfies all the requirements of Definition 2.2, $(W, +)$ is an abelian group and the scalar multiplication behaves according to the axioms. Thus, W is a vector space over $\mathbb{F}$. ■

Every vector space (non-zero) has two trivial subspaces, namely the zero subspace $\{0\}$ and the entire space V itself. We call these two subspaces the *trivial subspaces* of V . Any other subspace is called a *non-trivial subspace*. The first one of these, that is, $\{0\}$, is called the *zero subspace* of V .

Let us get back to the examples we had seen and write down some non-trivial subspaces of these examples.

(i) **n -tuple space:** For $V = \mathbb{F}^n$, some non-trivial subspaces include:

- The set of all vectors in $\mathbb{F}^n$ whose first coordinate is zero, that is,

$$\{(0, x_2, \dots, x_n) : x_2, \dots, x_n \in \mathbb{F}\}$$

- The set of all vectors in $\mathbb{F}^n$ whose first two coordinates are zero, that is,

$$\{(0, 0, x_3, \dots, x_n) : x_3, \dots, x_n \in \mathbb{F}\}$$

(ii) **Matrix Spaces:** For $V = M_{m \times n}(\mathbb{F})$, some non-trivial subspaces include:

- The set of all $m \times n$ matrices whose last column is zero, that is,

$$\{A \in M_{m \times n}(\mathbb{F}) : A_{i,n} = 0 \text{ for all } i\}$$

- The set of all $m \times n$ matrices whose (i_0, j_0) and (i_1, j_1) entries are zero for some fixed indices (i_0, j_0) and (i_1, j_1) , that is,

$$\{A \in M_{m \times n}(\mathbb{F}) : A_{i_0, j_0} = A_{i_1, j_1} = 0\}$$

(iii) **Field over field:** For $V = E$ where E is a field extension of F , some non-trivial subspaces include:

- The set of all elements in E that are fixed by a certain subgroup of the Galois group of E over F .
- The set of all elements in E that are algebraic over F .

(iv) **Polynomials over a field $\mathbb{F}$:** For $V = \mathbb{F}[x]_{\leq n}$, some non-trivial subspaces include:

- The set of all polynomials in $\mathbb{F}[x]_{\leq n}$ that have zero constant term, that is,

$$\{a_1x + a_2x^2 + \dots + a_nx^n : a_1, \dots, a_n \in \mathbb{F}\}$$

- The set of all polynomials in $\mathbb{F}[x]_{\leq n}$ that have zero linear term, that is,

$$\{a_0 + a_2x^2 + \dots + a_nx^n : a_0, a_2, \dots, a_n \in \mathbb{F}\}$$

Readers unfamiliar with Field extensions and Galois groups may ignore the examples of subspaces in the field over field case. Although I personally doubt anybody would read field and Galois theory without reading linear algebra first (although it is very much possible), I have included it for the sake of completeness.

We can perform many interesting operations on subspaces. We cover this soon but for now we talk about the intersection and union of subspaces. The intersection of two subspaces is again a subspace, but the union of two subspaces is not necessarily a subspace.

Proposition 2.8. *Let V be a vector space over a field $\mathbb{F}$ and let $W_1, W_2 \subseteq V$ be subspaces of V . Then the intersection $W_1 \cap W_2$ is also a subspace of V .*

Proof. We use the subspace criterion from Proposition 2.7 to show that $W_1 \cap W_2$ is a subspace of V .

First, we must ensure that $W_1 \cap W_2$ is non-empty. Since W_1 and W_2 are subspaces of V , both must contain the zero vector $0 \in V$ by Proposition 2.7. Therefore, $0 \in W_1 \cap W_2$, which confirms the intersection is non-empty.

Now, let $\lambda \in \mathbb{F}$ and let $u, v \in W_1 \cap W_2$. We need to show that $\lambda u + v \in W_1 \cap W_2$.

- (i) Since $u, v \in W_1 \cap W_2$, it follows by definition of the intersection that $u, v \in W_1$. Because W_1 is given to be a subspace of V , it is closed under linear combinations. Thus, $\lambda u + v \in W_1$.
- (ii) Similarly, since $u, v \in W_1 \cap W_2$, we also have $u, v \in W_2$. Because W_2 is a subspace of V , it is likewise closed under linear combinations. Thus, $\lambda u + v \in W_2$.

Since $\lambda u + v$ belongs to both W_1 and W_2 , it must belong to their intersection, meaning $\lambda u + v \in W_1 \cap W_2$.

Therefore, by Proposition 2.7, $W_1 \cap W_2$ is a subspace of V . ■

Question. Show that the union of two subspaces is not necessarily a subspace by giving an explicit example.

Ans. Consider the vector space $\mathbb{R}^2$ over the field $\mathbb{R}$. Let W_1 be the subspace of $\mathbb{R}^2$ consisting of all vectors of the form $(x, 0)$ where $x \in \mathbb{R}$, and let W_2 be the subspace of $\mathbb{R}^2$ consisting of all vectors of the form $(0, y)$ where $y \in \mathbb{R}$. Both W_1 and W_2 are subspaces of $\mathbb{R}^2$.

Now, consider the union $W_1 \cup W_2$. This set consists of all vectors that are either in W_1 or in W_2 , which means it includes all vectors of the form $(x, 0)$ and all vectors of the form $(0, y)$.

However, the union $W_1 \cup W_2$ is not closed under addition. For example, if we take the vector $(1, 0) \in W_1$ and the vector $(0, 1) \in W_2$, their sum is $(1, 1)$, which is not in either W_1 or W_2 , and hence not in their union.

Therefore, since $W_1 \cup W_2$ is not closed under addition, it cannot be a subspace of $\mathbb{R}^2$. This example demonstrates that the union of two subspaces is not necessarily a subspace. □

The reader is urged to come up with more examples of subspaces whose union is not a subspace. For instance, one can take the space of all polynomials of degree at most n and consider the subspace of all polynomials with zero constant term and the subspace of all polynomials with zero linear term. The union of these two subspaces is not a subspace.

However, we can improve upon this and ask what happens if the union is indeed a subspace. It turns out that in this case, one of the subspaces must be contained in the other.

Proof. Suppose that $W_1 \cup W_2$ is a subspace of V . Assume that $W_1 \not\subseteq W_2$. We must show that $W_2 \subseteq W_1$.

Let w be an arbitrary vector in W_2 . Since $W_1 \not\subseteq W_2$, there exists a vector $v \in W_1$ such that $v \notin W_2$.

Because $v \in W_1$ and $w \in W_2$, both vectors belong to the union $W_1 \cup W_2$. By our hypothesis, $W_1 \cup W_2$ is a subspace, so it is closed under vector addition. Therefore, the sum $v + w$ must also be in $W_1 \cup W_2$. This leaves us with two possibilities: either $v + w \in W_1$ or $v + w \in W_2$.

If $v + w \in W_2$, then since W_2 is a subspace, it must contain the additive inverse $-w$. This would imply that $(v + w) + (-w) = v \in W_2$, which directly contradicts our initial choice of $v \notin W_2$.

Thus, it must be the case that $v + w \in W_1$. Since W_1 is a subspace, it contains the additive inverse $-v$. Because W_1 is closed under addition, adding $-v$ to $v + w$ gives us $(-v) + (v + w) = w \in W_1$.

Since w was chosen as an arbitrary vector in W_2 , and we proved $w \in W_1$, we conclude that every vector in W_2 is also in W_1 . Therefore, $W_2 \subseteq W_1$. ■

We now discuss spans of vectors. The span of a set of vectors is the smallest subspace that contains those vectors, and it can be thought of as the set of all linear combinations of those vectors.

Definition 2.9. Let V be a vector space over a field $\mathbb{F}$ and let $S \subseteq V$ be a set of vectors in V . The **span**

of S , denoted $\text{span}\{S\}$, is the set of all linear combinations of vectors in S . Explicitly,

$$\text{span}\{S\} = \left\{ \sum_{i=1}^n c_i v_i \mid n \in \mathbb{N}, c_i \in \mathbb{F}, v_i \in S \right\}$$

Proposition 2.10. *Let V be a vector space over a field $\mathbb{F}$ and let $S \subseteq V$. Then $\text{span}\{S\}$ is a subspace of V . Moreover, it is the smallest subspace of V that contains S , in the sense that if W is any subspace of V such that $S \subseteq W$, then $\text{span}\{S\} \subseteq W$.*

Proof. First, we use Proposition 2.7 to show that $\text{span}\{S\}$ is a subspace of V . Suppose $x, y \in \text{span}\{S\}$. Then x and y can be written as finite linear combinations of vectors in S . That is, there exist scalars $a_1, \dots, a_n, b_1, \dots, b_m \in \mathbb{F}$ and vectors $u_1, \dots, u_n, w_1, \dots, w_m \in S$ such that

$$x = \sum_{i=1}^n a_i u_i \quad \text{and} \quad y = \sum_{j=1}^m b_j w_j$$

We must show that $\text{span}\{S\}$ is closed under addition and scalar multiplication.

(i) For addition, we have

$$x + y = \sum_{i=1}^n a_i u_i + \sum_{j=1}^m b_j w_j$$

This is simply a finite sum of $n + m$ terms, where each term is a scalar multiple of a vector in S . Therefore, $x + y$ is a linear combination of vectors in S , which means $x + y \in \text{span}\{S\}$.

(ii) For scalar multiplication, let $\lambda \in \mathbb{F}$. Then

$$\lambda x = \lambda \left(\sum_{i=1}^n a_i u_i \right) = \sum_{i=1}^n (\lambda a_i) u_i$$

Since $\lambda a_i \in \mathbb{F}$ for each i , this is also a finite linear combination of vectors in S . Thus, $\lambda x \in \text{span}\{S\}$.

Since $\text{span}\{S\}$ is closed under both vector addition and scalar multiplication, it is a subspace of V .

Next, we show that $\text{span}\{S\}$ is the smallest subspace containing S . First, observe that $S \subseteq \text{span}\{S\}$. This is because for any $v \in S$, we can write $v = 1v$, which is a valid linear combination of elements in S .

Now, let W be any arbitrary subspace of V such that $S \subseteq W$. We need to show that $\text{span}\{S\} \subseteq W$. Let $v \in \text{span}\{S\}$. By Definition 2.9, we can write

$$v = \sum_{i=1}^k c_i v_i$$

for some scalars $c_i \in \mathbb{F}$ and vectors $v_i \in S$. Since $S \subseteq W$, it follows that $v_1, \dots, v_k \in W$. Because W is a subspace, it must be closed under scalar multiplication, meaning $c_i v_i \in W$ for all i . Furthermore, since W is closed under vector addition, the finite sum of these elements must also reside in W . Therefore, $v \in W$.

Since this holds for every $v \in \text{span}\{S\}$, we have $\text{span}\{S\} \subseteq W$ which shows that $\text{span}\{S\}$ is the smallest subspace of V containing S . ■

Proposition 2.11. *Let V be a vector space over a field $\mathbb{F}$ and let S be a subset of V . Then $\text{span}\{S\}$ is equal to the intersection of all subspaces of V that contain S .*

Proof. Let W be the intersection of all subspaces of V that contain S . We want to show that $\text{span}\{S\} = W$. We will do this by showing mutual inclusion.

First, $\text{span}\{S\}$ is a subset of any subspace of V that contains S from the above proposition. This shows that $\text{span}\{S\}$ is contained in the intersection of all such subspaces, which means $\text{span}\{S\} \subseteq W$.

Conversely, we know that $\text{span}\{S\}$ is itself a subspace of V that contains S . Because W is the intersection of *all* such subspaces, W must be contained in any particular one of them, including $\text{span}\{S\}$. Thus, $W \subseteq \text{span}\{S\}$. ■

Proposition 2.12. *Let V be a vector space over a field $\mathbb{F}$ and let $S, T \subseteq V$ be subsets of V . If $S \subseteq T$, then $\text{span}\{S\} \subseteq \text{span}\{T\}$.*

Proof. Let $v \in \text{span}\{S\}$. This means v can be expressed as a finite linear combination of vectors in S . That is, there exist vectors $s_1, \dots, s_n \in S$ and scalars $c_1, \dots, c_n \in \mathbb{F}$ such that

$$v = \sum_{i=1}^n c_i s_i$$

Since we are given that $S \subseteq T$, every vector $s_i \in S$ is also a vector in T . Consequently, the expression $\sum_{i=1}^n c_i s_i$ is also a valid finite linear combination of vectors in T .

Since v can be written as a linear combination of elements in T , it must belong to the span of T . Therefore, $v \in \text{span}\{T\}$. Because this holds for an arbitrary $v \in \text{span}\{S\}$, we conclude that $\text{span}\{S\} \subseteq \text{span}\{T\}$. ■

We now discuss some important operations on subspaces, namely the sum and direct sum of subspaces. We also discuss the notion of a quotient space, which is an important construction in linear algebra.

Definition 2.13. Let V be a vector space over a field $\mathbb{F}$ and let $U, W \subseteq V$ be subspaces of V . The **sum** of U and W , denoted by $U + W$, is the set of all vectors of the form $u + w$ where $u \in U$ and $w \in W$. Explicitly,

$$U + W = \{u + w : u \in U, w \in W\}$$

Proposition 2.14. *Let V be a vector space over a field $\mathbb{F}$ and let $U, W \subseteq V$ be subspaces of V . Then the sum $U + W$ is also a subspace of V .*

Proof. We use Proposition 2.7 to show that $U + W$ is a subspace of V . Let $\lambda \in \mathbb{F}$ and let $x, y \in U + W$. Then there exist $u_1, u_2 \in U$ and $w_1, w_2 \in W$ such that $x = u_1 + w_1$ and $y = u_2 + w_2$. We need to show that $\lambda x + y \in U + W$.

$$\begin{aligned} \lambda x + y &= \lambda(u_1 + w_1) + (u_2 + w_2) \\ &= (\lambda u_1 + u_2) + (\lambda w_1 + w_2) \end{aligned}$$

Since U is a subspace, it is closed under linear combinations, so $\lambda u_1 + u_2 \in U$. Similarly, since W is a subspace, $\lambda w_1 + w_2 \in W$. Thus, we can write $\lambda x + y$ as the sum of an element in U and an element in W , which means $\lambda x + y \in U + W$. ■

Proposition 2.15. *Let V be a vector space over a field $\mathbb{F}$ and let $U, W \subseteq V$ be subspaces of V . Then $U + W = \text{span}\{U \cup W\}$.*

Proof. We first show that $U + W \subseteq \text{span}\{U \cup W\}$. Let $x \in U + W$. Then there exist $u \in U$ and $w \in W$ such that $x = u + w$. Since $u \in U$ and $w \in W$, both u and w are elements of $U \cup W$. Therefore, x can be expressed as a linear combination of elements in $U \cup W$ (specifically, as the sum of two elements), which shows $x \in \text{span}\{U \cup W\}$.

Next, we show that $\text{span}\{U \cup W\} \subseteq U + W$. Let $y \in \text{span}\{U \cup W\}$. By definition of span, there exist vectors $v_1, v_2, \dots, v_n \in U \cup W$ and scalars $c_1, c_2, \dots, c_n \in \mathbb{F}$ such that

$$y = c_1 v_1 + c_2 v_2 + \dots + c_n v_n$$

Each vector v_i is either in U or in W . We can group the terms in the linear combination according to whether they belong to U or to W . Let $u = c_{i_1} v_{i_1} + c_{i_2} v_{i_2} + \dots + c_{i_k} v_{i_k}$ be the sum of all terms where $v_{i_j} \in U$ and

let $w = c_{j_1}v_{j_1} + c_{j_2}v_{j_2} + \cdots + c_{j_m}v_{j_m}$ be the sum of all terms where $v_{j_i} \in W$. Since U and W are subspaces, both u and w are elements of their respective subspaces. Thus, we can write

$$y = u + w$$

where $u \in U$ and $w \in W$. This shows that $y \in U + W$. ■

We now have the notion of a direct sum of subspaces, which is a special case of the sum of subspaces where the intersection is trivial.

Definition 2.16. Let V be a vector space over a field $\mathbb{F}$ and let $U, W \subseteq V$ be subspaces of V . The **direct sum** of U and W , denoted by $U \oplus W$, is the sum $U + W$ together with the condition that $U \cap W = \{0\}$.

Direct sums have the following important property: every element in the direct sum can be uniquely expressed as a sum of an element from U and an element from W .

Proposition 2.17. Let V be a vector space over a field $\mathbb{F}$ and let $U, W \subseteq V$ be subspaces of V . If $V = U \oplus W$, then every element $v \in V$ can be uniquely written as $v = u + w$ where $u \in U$ and $w \in W$.

Proof. Suppose there are two representations of $v \in V$, say $v = u_1 + w_1$ and $v = u_2 + w_2$ where $u_1, u_2 \in U$ and $w_1, w_2 \in W$. Then we have

$$u_1 + w_1 = u_2 + w_2$$

Rearranging this gives

$$(u_1 - u_2) = (w_2 - w_1)$$

The left-hand side of this equation is an element of U , while the right-hand side is an element of W . Since $U \cap W = \{0\}$, the only way for these two sides to be equal is if both sides are equal to the zero vector. Thus, we have

$$u_1 - u_2 = 0 \quad \text{and} \quad w_2 - w_1 = 0$$

This implies that $u_1 = u_2$ and $w_1 = w_2$. Therefore, the representation of v as a sum of an element from U and an element from W is unique. ■

We now define the notion of a quotient space, which is an important construction in linear algebra.

Definition 2.18. Let V be a vector space over a field $\mathbb{F}$ and let $W \subseteq V$ be a subspace of V . The **quotient space** of V by W , denoted by V/W , is the set of equivalence classes of vectors in V under the equivalence relation defined by $v \sim u$ if and only if $v - u \in W$. The elements of V/W are called **cosets** of W in V .

The readers are invited to check that the given relation is indeed an equivalence relation. Note that we are calling the quotient set a “space”. This is because it is indeed a vector space over $\mathbb{F}$. Firstly, let the equivalence class of a vector $v \in V$ be denoted by $[v]$. We can define addition and scalar multiplication on V/W as follows:

$$\begin{aligned} [v] + [u] &= [v + u] \\ \lambda[v] &= [\lambda v] \end{aligned}$$

Proposition 2.19. Let V be a vector space over a field $\mathbb{F}$ and let $W \subseteq V$ be a subspace of V . The operations on V/W are well-defined. Furthermore, the quotient space V/W is a vector space over $\mathbb{F}$ under these operations.

Proof. First, we check that addition is well-defined. Suppose $[v] = [v']$ and $[u] = [u']$, which means $v - v' \in W$ and $u - u' \in W$. We need to show that $[v + u] = [v' + u']$. Consider the difference:

$$(v + u) - (v' + u') = (v - v') + (u - u')$$

Since both $v - v'$ and $u - u'$ are elements of W , their sum is also an element of W . Therefore, $(v + u) - (v' + u') \in W$, which implies that $[v + u] = [v' + u']$. Thus, addition is well-defined.

Next, we check that scalar multiplication is well-defined. Suppose $[v] = [v']$, which means $v - v' \in W$. We need to show that $\lambda[v] = \lambda[v']$, or equivalently, that $[\lambda v] = [\lambda v']$. Consider the difference:

$$\lambda v - \lambda v' = \lambda(v - v')$$

Since $v - v' \in W$ and W is a subspace, it is closed under scalar multiplication. Therefore, $\lambda(v - v') \in W$, which implies that $[\lambda v] = [\lambda v']$. Thus, scalar multiplication is well-defined.

Now that we have established that the operations are well-defined, we can verify the axioms of a vector space for V/W .

(i) Axioms for $(V/W, +)$ being an abelian group:

(a) Associativity: For all $[u], [v], [w] \in V/W$,

$$([u] + [v]) + [w] = [u + v] + [w] = [(u + v) + w] = [u + (v + w)] = [u] + [v + w] = [u] + ([v] + [w])$$

(b) Commutativity: For all $[u], [v] \in V/W$,

$$[u] + [v] = [u + v] = [v + u] = [v] + [u]$$

(c) Identity element: The zero vector in V/W is the equivalence class $[0] = W$. Indeed, for all $[v] \in V/W$,

$$[v] + [0] = [v + 0] = [v]$$

(d) Inverse element: For every $[v] \in V/W$, its additive inverse is $[-v]$. Indeed,

$$[v] + [-v] = [v + (-v)] = [0]$$

(ii) Axioms for scalar multiplication:

(a) Distributivity over vector addition: For all $\lambda \in \mathbb{F}$ and $[u], [v] \in V/W$,

$$\lambda([u] + [v]) = \lambda[u + v] = [\lambda(u + v)] = [\lambda u + \lambda v] = [\lambda u] + [\lambda v] = \lambda[u] + \lambda[v]$$

(b) Distributivity over scalar addition: For all $\lambda, \mu \in \mathbb{F}$ and $[v] \in V/W$,

$$(\lambda + \mu)[v] = [(\lambda + \mu)v] = [\lambda v + \mu v] = [\lambda v] + [\mu v] = \lambda[v] + \mu[v]$$

(c) Associativity of scalar multiplication: For all $\lambda, \mu \in \mathbb{F}$ and $[v] \in V/W$,

$$(\lambda\mu)[v] = [(\lambda\mu)v] = [\lambda(\mu v)] = \lambda[\mu v] = \lambda(\mu[v])$$

(d) Multiplicative identity: For all $[v] \in V/W$,

$$1[v] = [1v] = [v]$$

Since all eight axioms are satisfied, the quotient space V/W is a vector space over $\mathbb{F}$. ■

There are many important properties of quotient spaces. For instance, there is a natural linear map from V to V/W that sends each vector to its equivalence class, and the kernel of this map is precisely the subspace W . This leads to the important isomorphism theorems in linear algebra. The reader is invited to revisit this section after learning about linear maps and isomorphisms to see these connections.

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